

Meeting – July 9

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Overview

- Link Quality Analysis
- Leveraging MeshMonitor to observe mesh behavior in real-time
- Dynamic/Moving Node testing

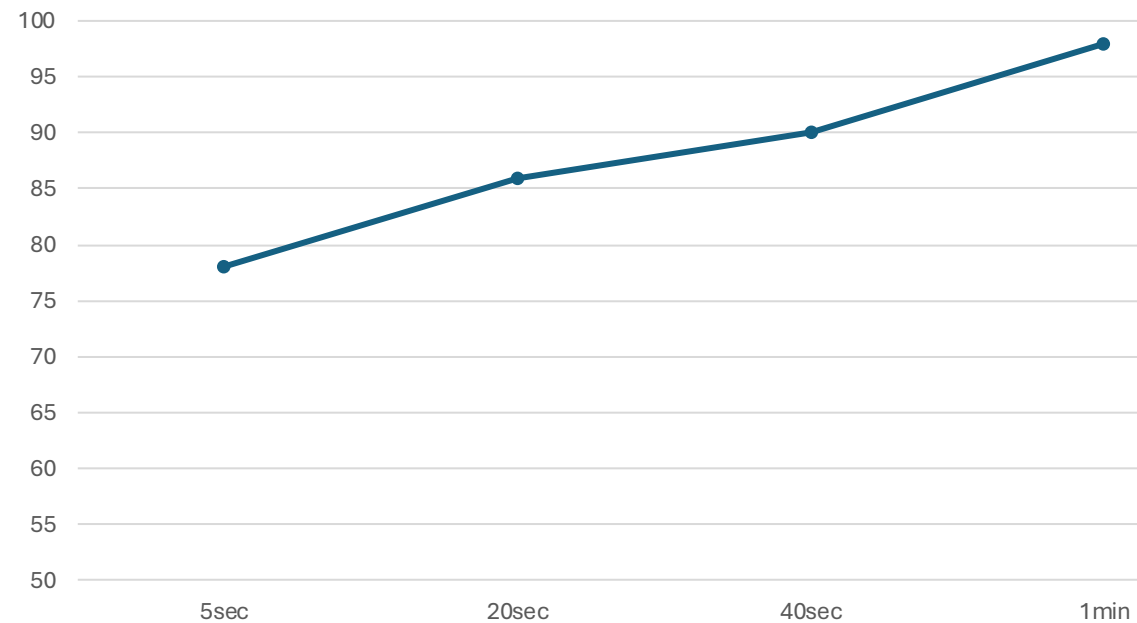
Link Quality Analysis

- Earlier this week, I continued the link quality analysis, running the same tests as before except at a different time of day.
- Additionally, I made sure to include both sender and receiver timestamps to calculate end-to end latency.
- The results for the afternoon tests were, for the most part, similar to those from the morning.
- Some notable observations include:
 - The delivery rate 20 and 40 second intervals was lower on the afternoon test, but maintained the same increasing trend
 - The 5 second interval had high latency, with packets being received in a different order than they were sent and a large variance in latency values.

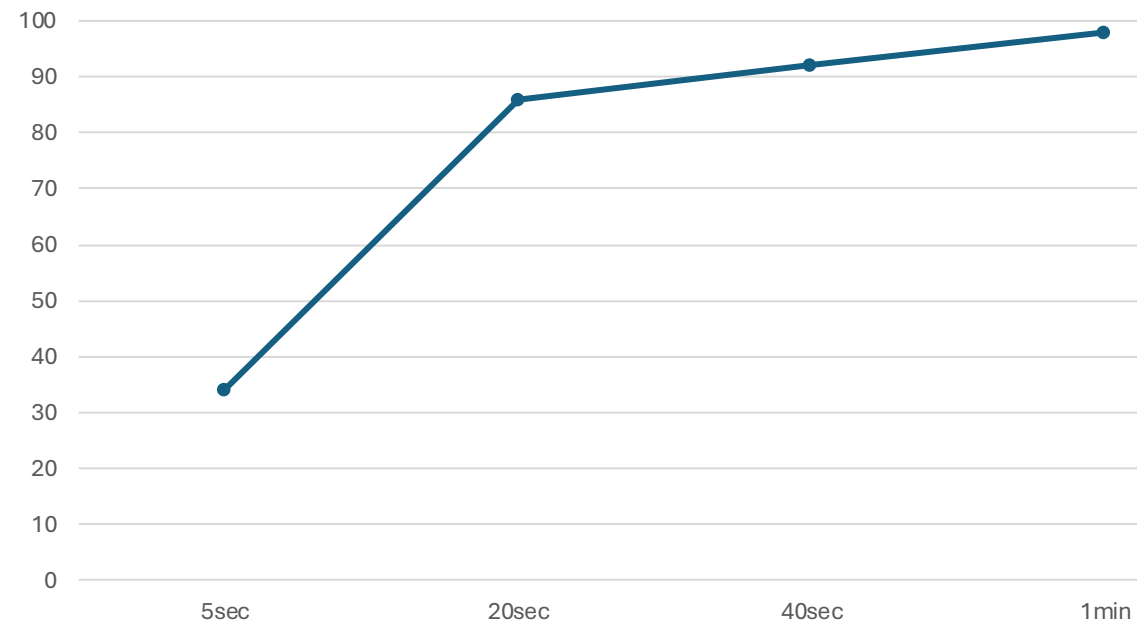
AFTERNOON	5sec	20sec	40sec	1min	total	
DR (%)	78	86	90	98	88.5	
latency	65.9918098	6.23665905	6.61673935	6.21463817	6.35601219	21.2649616
latency SD	44.0988802	0.81603368	1.75006418	0.84907817	1.22359576	32.1789572
avg hops	1.8974359	1.97674419	1.95652174	2	1.95767546	
avg RSSI (dBm, outgoing)	-138.17949	-138.60465	-138.8913	-137.4898	-138.29131	
RSSI SD (outgoing)	1.76026007	1.77468354	2.19320426	1.20972533	1.83004444	
avg SNR (dB, outgoing)	-14.121795	-13.976744	-14.440217	-12.877551	-13.854077	
SNR SD (outgoing)	1.70223517	1.67688589	2.05763131	1.16046243	1.76647379	
ACKs received	17	43	46	49	155	
ACKs received (%)	34	86	92	98	77.5	
avg RSSI (dBm, ACK)	-118	-121.37209	-118.77778	-120.7551	-120.1	
RSSI SD (ACK)	7.85281266	10.9870557	10.3218411	11.7287585	10.7838744	
avg SNR (dB, ACK)	2.96153846	0.98837209	2.77777778	1.10714286	1.735	
SNR SD (ACK)	6.73597784	9.50907359	8.93381953	10.2525403	9.34473212	

Cells highlighted in orange have extreme, unusual values. The average for these do not take these into account, unless also highlighted in orange.

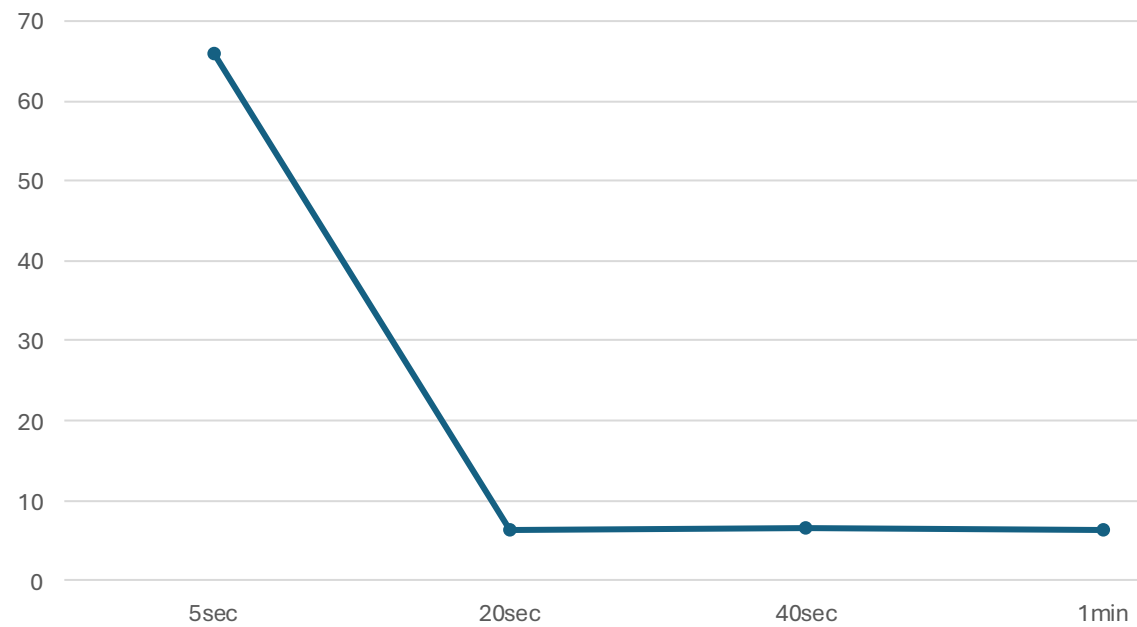
Afternoon - DR (%)



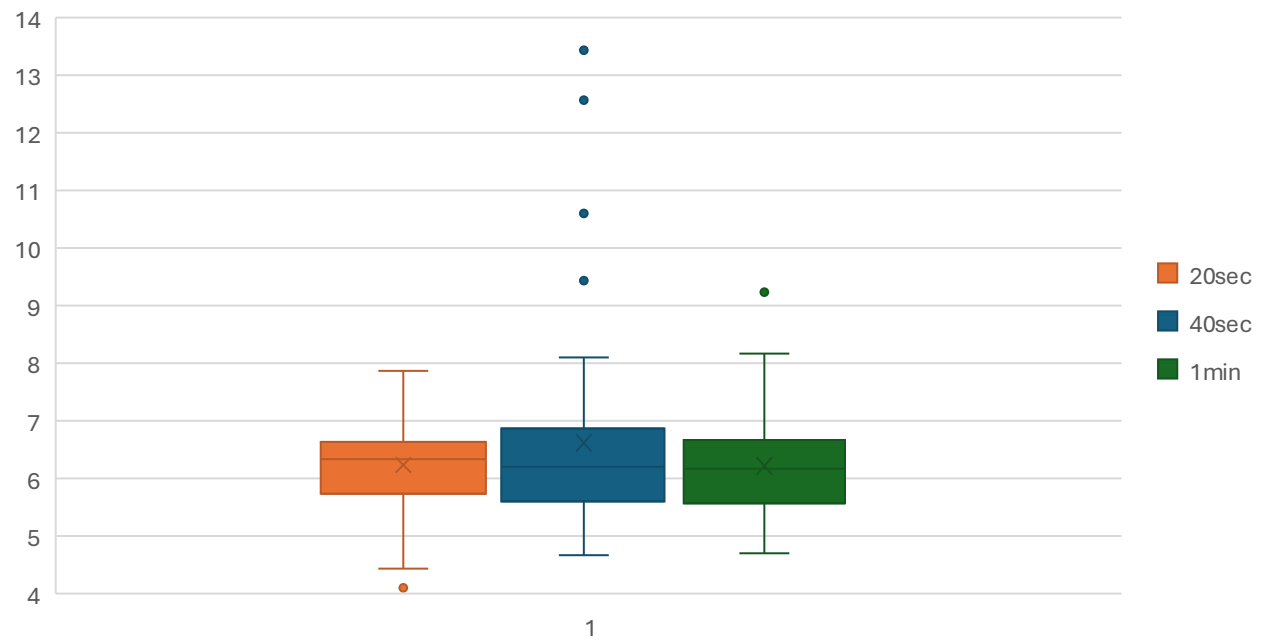
Afternoon - ACKs received (%)



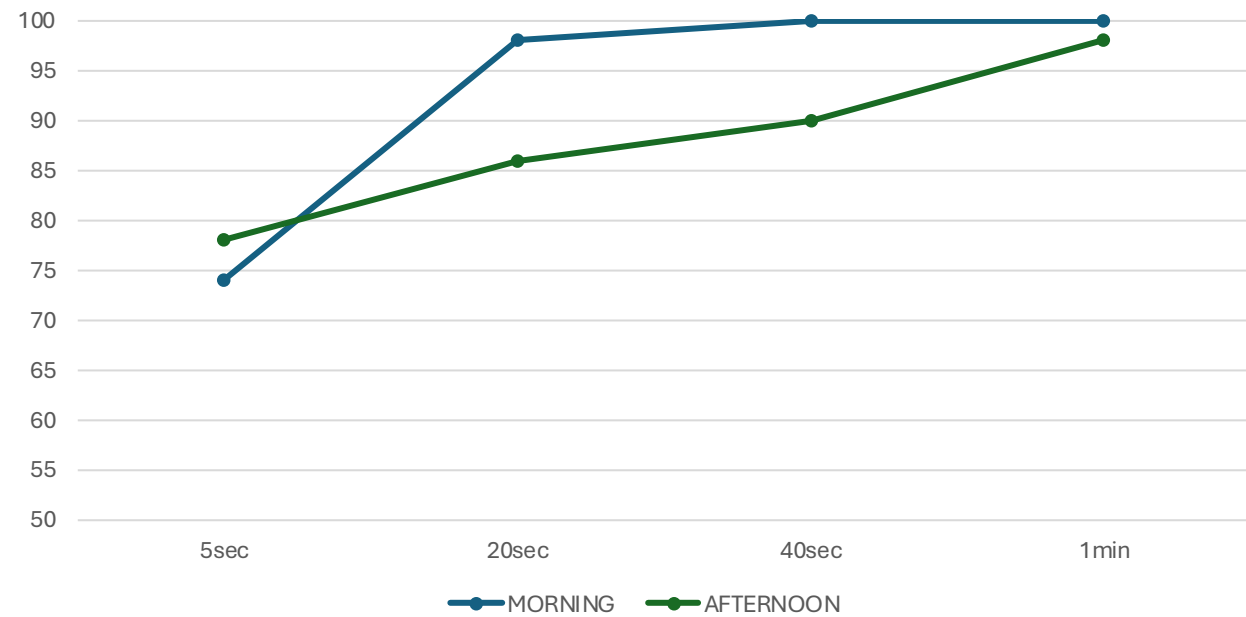
Afternoon - Latency (avg)



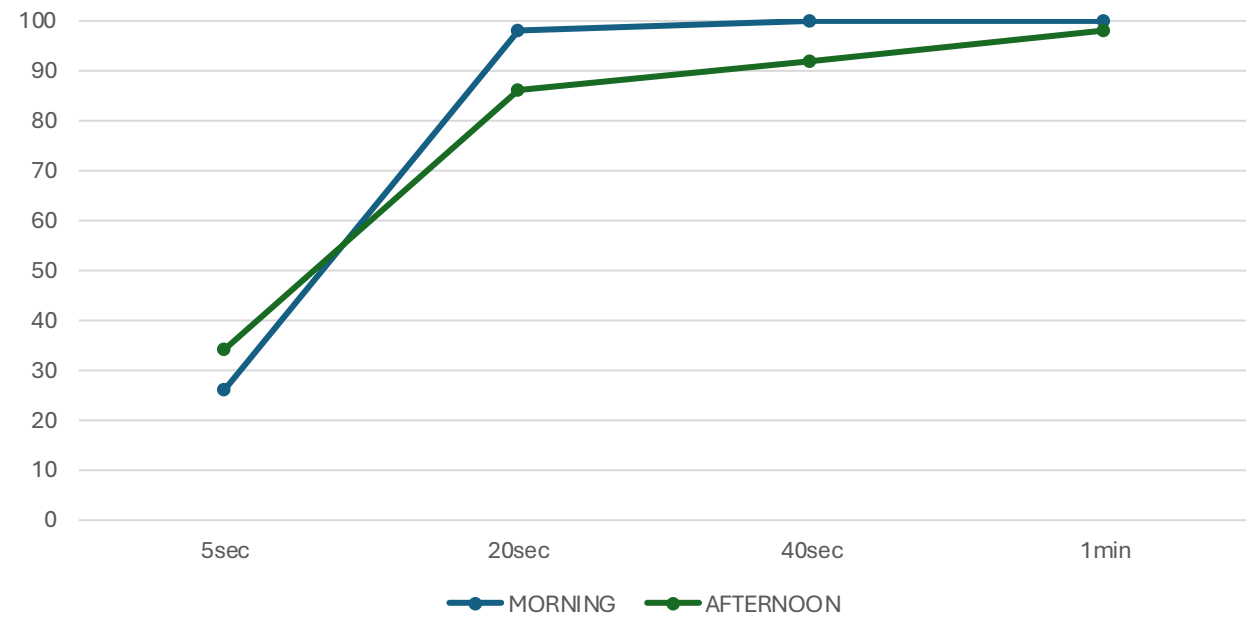
Afternoon - Latency



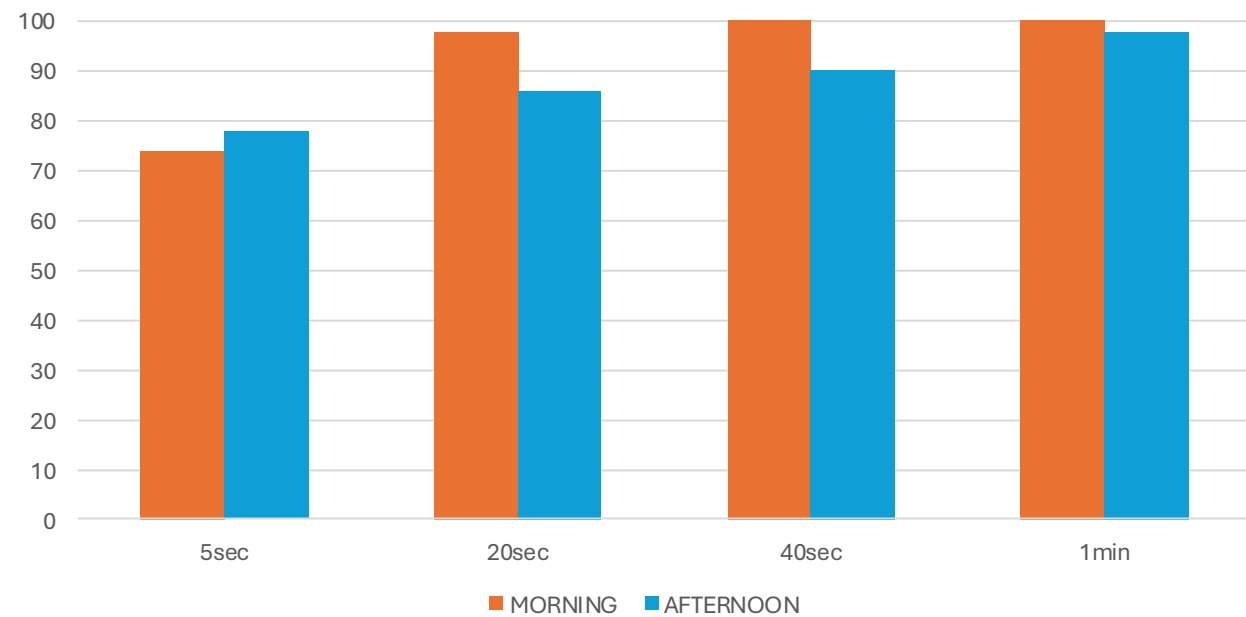
All Times - DR%



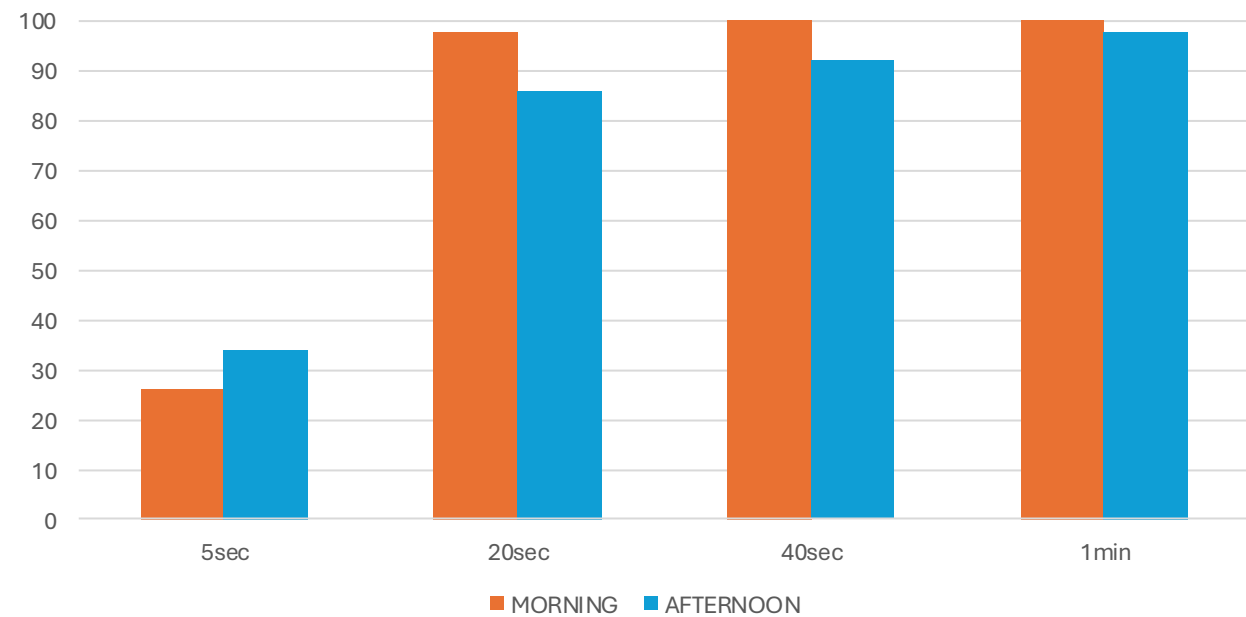
All Times - ACKs Received per Packet Sent (%)



All Times - DR%



All Times - ACKs Received per Packet Sent (%)



Leveraging MeshMonitor

- While MeshMonitor is a fantastic tool for remotely monitoring, tweaking and updating LoRa mesh nodes, it lacked one function I was looking for: to observe how often **all** nodes were transmitting and receiving packets.
- MeshMonitor by default exposes this data for the local node, but not for other remote nodes.
- However, MeshMonitor does have a tool which automates local_stats telemetry request from certain nodes at a given interval.
- Using this tool, I was able to create a live Grafana dashboard to monitor in real-time which nodes are transmitting the most, relaying and receiving the most packets

Dashboard Snapshot

- Link:

<https://snapshots.raintank.io/dashboard/snapshot/CA5L7yuOC61DByQpqEJKmOBDbEw8ReeN>

Meshtastic Dashboard

- All the dashboard metrics are measured as a rate of:
 - $\frac{\text{Packets since last transmission}}{\text{Minutes since last transmission}}$ i.e., Packets per Minute
- While the dashboard is live, the link sent is only a snapshot of the live feed. If the time permits, it would be a good idea to host the dashboard publicly, allowing for the remote monitoring of the mesh from any device.
- All the collected local_stats telemetry data used for the dashboard is stored locally in a SQLite database.
 - In fact, all data collected by MeshMonitor is stored in this database.

Dynamic Node Testing

- This morning, I ran a test where I walked around campus sending packets from a moving node and receiving them here at the Goldberg.
- This test was quite short. Because so, I believe it would be nice to run this test multiple times to gather more data for stronger insights.
- Next time, it would be good to track how many packets are being sent for each number of hops, showing how drastically delivery rate changes as there are more retransmissions.

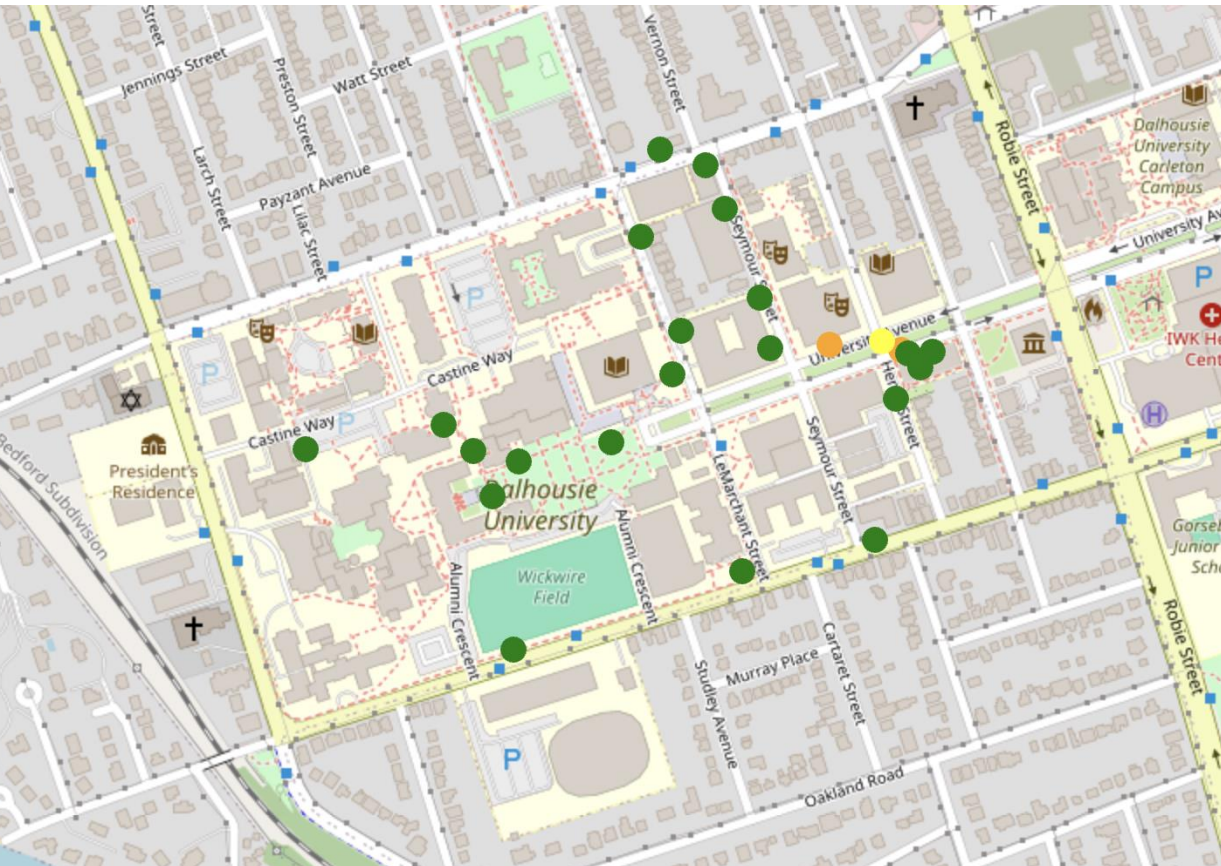
localID	packetID	nodeID	sendTime	receivedTime	Latency(s)	rssI (dBm)	snr (dB)	lat	long	alt	hopStart	hopLimit	hopsUsed	delivery rate (%)
1	2633666616	!dadfb03c	1783607744.8772821	1783607747.1792386	2.301949978	-39	7.75	44.637543	-63.587178	48	3	3	0	50
2	3477515321	!dadfb03c	1783607774.996068	1783607777.1019025	2.105840206	-51	7.5	44.637543	-63.587178	48	3	3	0	
3	2254213178	!dadfb03c	1783607805.116203	1783607807.083689	1.967479944	-43	7.5	44.637543	-63.587178	48	3	3	0	
5	855942204	!dadfb03c	1783607865.351221	1783607867.265962	1.914740086	-110	-0.5	44.63756	-63.587568	50	3	3	0	
6	308669501	!dadfb03c	1783607895.464409	1783607900.216167	4.751760006	-78	9.25	44.63713	-63.587627	48	3	2	1	
10	2674245697	!dadfb03c	1783608015.924809	1783608020.124709	4.19990015	-78	9	44.635899	-63.587878	46	3	2	1	
13	1213493316	!dadfb03c	1783608106.2908878	1783608113.5910997	7.300209999	-80	8.75	44.635615	-63.589516	49	3	1	2	
18	775394377	!dadfb03c	1783608256.9066129	1783608265.131039	8.224420071	-80	7.75	44.634939	-63.592325	44	3		3	
27	2458402898	!dadfb03c	1783608528.0274801	1783608535.955114	7.927630186	-78	8.75	44.636687	-63.594874	46	3		3	
31	2914107478	!dadfb03c	1783608648.514365	1783608657.2877207	8.773360014	-84	8.75	44.636906	-63.593182	53	3		3	
32	3886922839	!dadfb03c	1783608678.640099	1783608685.6454601	7.005369902	-81	8.5	44.636679	-63.592816	52	3	1	2	
33	1128078424	!dadfb03c	1783608708.768106	1783608712.8818574	4.113749981	-80	9	44.636278	-63.592574	52	3	2	1	
34	3675344985	!dadfb03c	1783608738.896904	1783608743.2310798	4.334170103	-79	8.75	44.636584	-63.592254	51	3	2	1	
36	2969691	!dadfb03c	1783608799.151428	1783608802.8935812	3.742159843	-79	9.75	44.636744	-63.591111	50	3	2	1	
38	1666175069	!dadfb03c	1783608859.406923	1783608869.445931	10.03901005	-79	9.25	44.637351	-63.590374	54	3	1	2	
39	1819711582	!dadfb03c	1783608889.534632	1783608896.2272413	6.692610025	-80	7.75	44.63772	-63.590268	55	3	1	2	
41	2244295776	!dadfb03c	1783608949.779661	1783608953.607958	3.828289986	-81	8.75	44.638552	-63.59076	56	3	2	1	
43	876472418	!dadfb03c	1783609010.028699	1783609019.4571238	9.428429842	-81	8.75	44.639318	-63.590517	57	3		3	
44	2327524451	!dadfb03c	1783609040.147904	1783609048.4374907	8.289589882	-79	9.25	44.639173	-63.58997	58	3	1	2	
45	847907940	!dadfb03c	1783609070.272023	1783609074.4896786	4.217649937	-81	9	44.6388	-63.589725	57	3	2	1	
47	2635497574	!dadfb03c	1783609130.5294101	1783609134.5983536	4.068940163	-81	8	44.638023	-63.589286	55	3	2	1	
48	2951375975	!dadfb03c	1783609160.654915	1783609164.9157374	4.260819912	-78	9.75	44.637579	-63.589177	54	3	2	1	
49	4144805992	!dadfb03c	1783609190.782335	1783609192.5468304	1.764499903	-112	-1.25	44.637602	-63.588443	52	3	3	0	
50	2828412009	!dadfb03c	1783609220.9005299	1783609222.8061585	1.905629873	-102	6.5	44.637636	-63.587785	50	3	3	0	
51	3642589290	!dadfb03c	1783609251.024917	1783609252.796444	1.771529913	-89	8	44.637533	-63.587479	48	3	3	0	

Green: > -90 dBm
Yellow: -110 to -90 dBm
Orange: -130 to -110 dBm
Red: ≤ -130 dBm

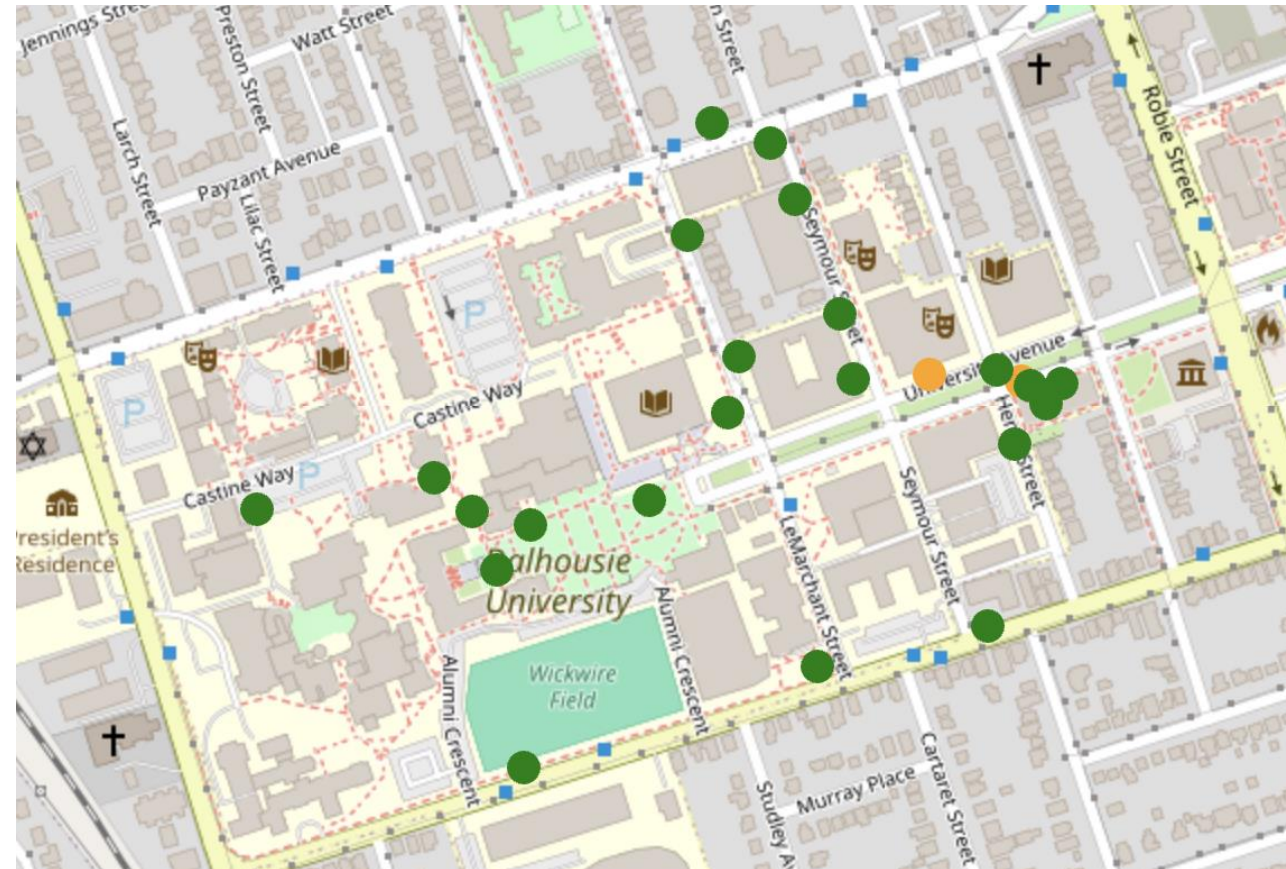
Dynamic Node Testing

Dark Green: > 10 dB
Green: 5 to 10 dB
Yellow: 0 to 5 dB
Orange: -5 to 0 dB
Red: -10 to -5 dB
Dark Red: ≤ -10 dB

RSSI



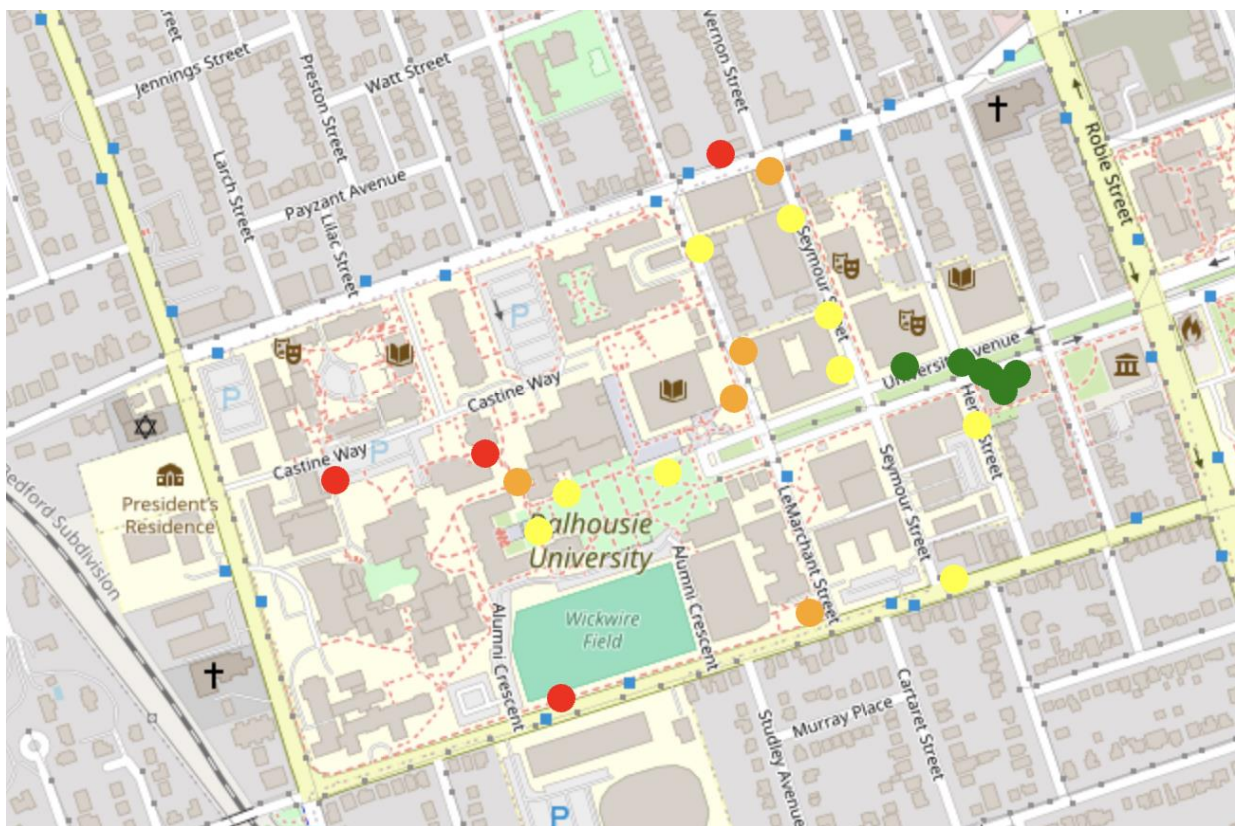
SNR



Dynamic Node Testing

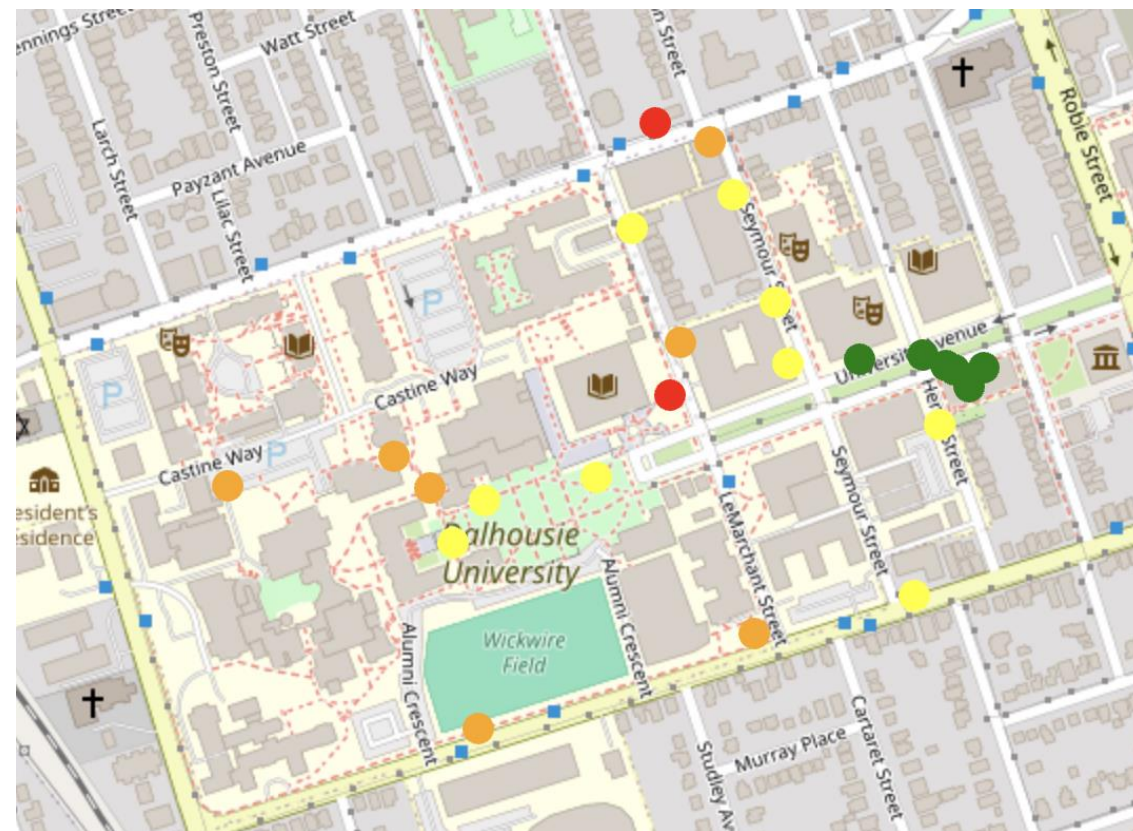
Green: 0 hops
Yellow: 1 hop
Orange: 2 hops
Red: 3+ hops

Hops



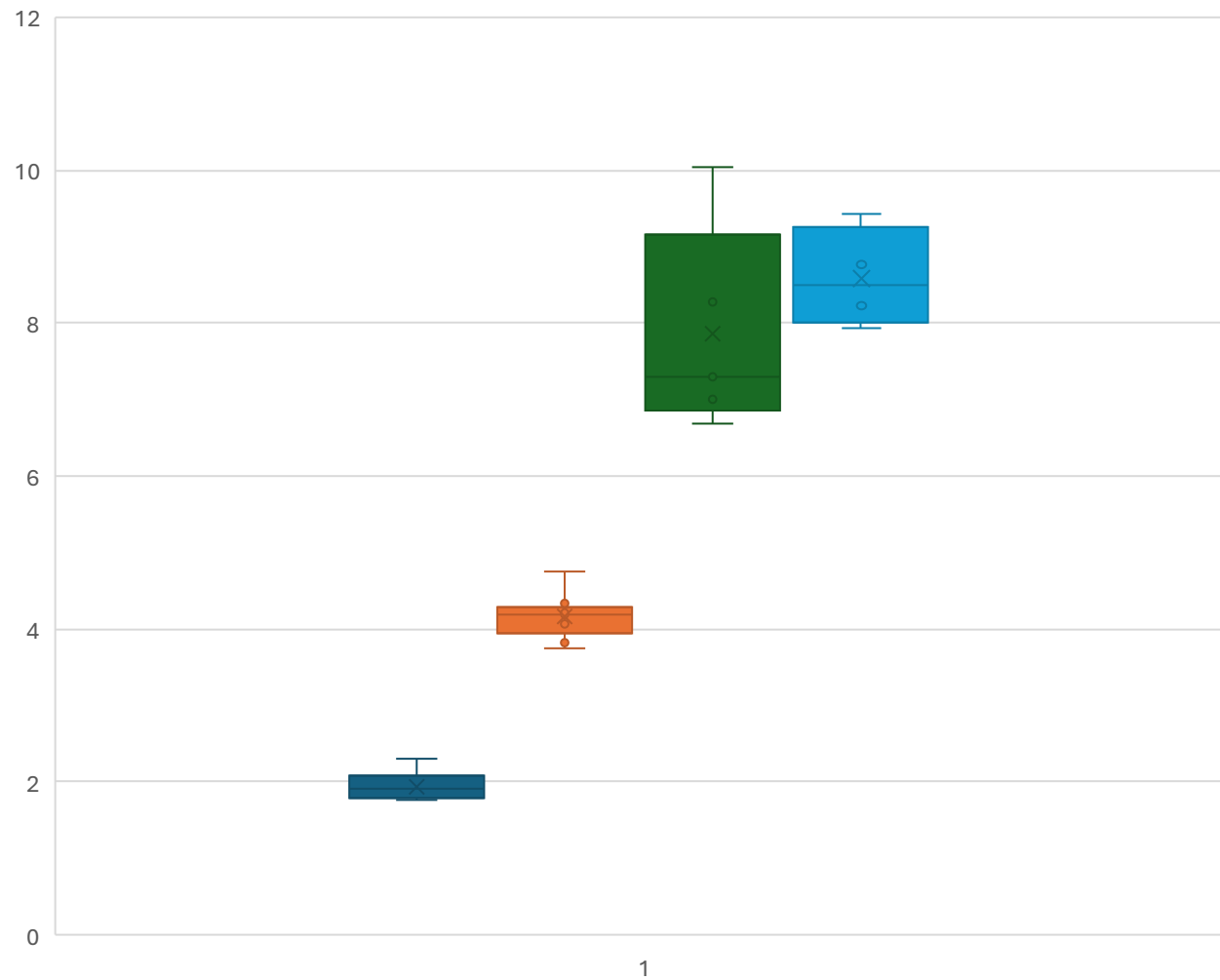
Green: < 3 s
Yellow: 3 to < 6 s
Orange: 6 to < 9 s
Red: ≥ 9 s

Latency



Dynamic Node Testing

Latency per Hop (s)



avg. latency by hops used (seconds)

